

PAPER_1671_ROOM_TEMP_SC_500_K

Daniel T. Murphy

PAPER_1671 — Room-T Superconductor Ceiling = $T_{c_max} = HTSC \times D_{phys} = 125 \times 4 = 500 \text{ K} = 227^\circ\text{C}$

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Superconductivity

Date: June 18, 2026

Status: CLOSED — EXACT closure (PAPER_136x broader corpus)

Observation

PAPER_1367 (PAPER_136x broader corpus) gives:

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$$T_{c_max} = HTSC \times D_{phys} = 125 \times 4 = 500 \text{ K} = 227^\circ\text{C}$$

``

Status: **EXACT**.

UQFF Closed Identity

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$$T_{c_max} = HTSC \times D_{phys} = 125 \times 4 = 500 \text{ K} = 227^\circ\text{C} \text{ EXACT}$$

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Room-Temperature Superconductor Ceiling

The upper bound on superconducting transition temperatures at ambient pressure:

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$$T_{c_max} = HTSC \times D_{phys} = 125 \text{ K} \times 4 = 500 \text{ K} = 227^\circ\text{C}$$

``

The ceiling is 500 K (227°C), comfortably above room temperature (300 K). UQFF predicts:

- Room-T SC is **achievable in principle** but requires reaching the integer-primitive bound
- Current high-pressure RTS materials (H $\blacksquare$ S 203 K, LaH $\blacksquare\blacksquare$ ~250 K, C-S-H disputed ~287 K) are below the UQFF ceiling but on a clear approach trajectory
- The 4 $\times$ multiplier from the base HTSC (125 K via PAPER_1659) is precisely the spacetime dimension D_{phys}

Practical implication: Material engineers targeting 500 K SC have a theoretical maximum to optimize toward, derived from UQFF integer primitives rather than empirical extrapolation.

NOT REPLACEMENT

Empirical superconductivity measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1367
- Related: PAPER_1656-1665 (tier-24); PAPER_1646-1655 (tier-23); PAPER_1636-1645 (tier-22)
- Calculator dispatch: `calculate_paradox({"paradox": "room_temp_sc_500_k"})`

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